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# Smart Musical Bell That Identifies The Person

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Musical bell is the choice of many over ding-dong type bell or door-lens for which you have to go to the door to look out. An IoT based smart musical bell can be rung through the conventional bell switch as well as by a motion detector sensor like an infrared (IR) sensor, passive infrared (PIR) sensor, or ultrasonic sensor, and even through a mobile phone.

Members of a family can use a smartphone application providing separate programmable tunes so that those at home can identify which member is ringing the doorbell on hearing the tune. Separate tunes can be set for different members of the family, and for guests and others a unique tune can be generated through a sensor. This musical bell can also be used in an office as a call bell that is operated through the executives' smartphones.

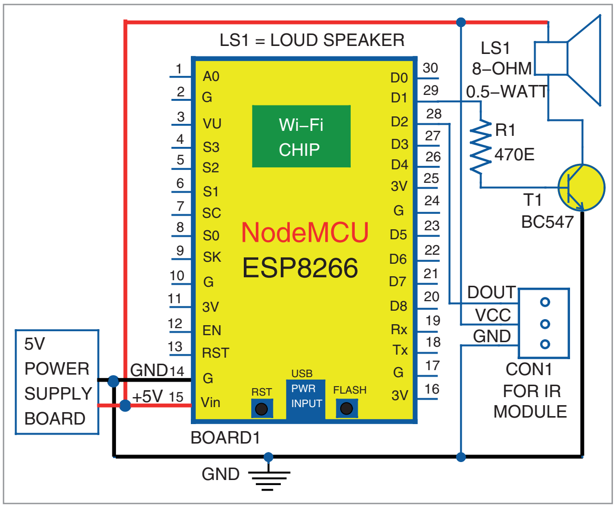


*Fig. 1: Circuit diagram of programmable smart musical bell*

In this low-cost programmable bell an IR sensor is used to detect the presence of person outside the door but you can also use a PIR sensor or an ultrasonic sensor.

## Circuit and description

Circuit diagram of the smart musical bell is shown in Fig. 1. It comprises NodeMCU board Board1, an IR sensor module, BC547 transistor T1, loudspeaker LS1, and 5V DC power supply. The NodeMCU is a single-board microcontroller that uses Wi-Fi chip ESP8266 meant for IoT applications. Its open source hardware and firmware uses Lua scripting language. The ESP-12 Node MCU used here has following features:

- An analogue input (10-bit integrated ADC)
- 13 digital I/O pins (alternate functions like USART/I2C/SPI)
- Micro USB port for power, programming, and debugging
- Two 2.54mm, 15-pin headers with access to GPIOs, SPI, UART, ADC, and power pins
- Reset and Flash buttons
- 5V power supply via micro USB port
- Wi-Fi support (802.11 b/g/n, 2.4 GHz, WPA/WPA2)
- Integrated low-power 32-bit RISC MCU (L106)
- 64kB instruction RAM, 96kB data RAM
- Integrated TCP/IP protocol stack
- Integrated TR switch, balun, LNA, power amplifier, and matching network
- Integrated PLL, regulators, and power management units
- Antenna diversity support
- Standby power consumption of <1.0mW
- +20dBm output power in 802.11b mode

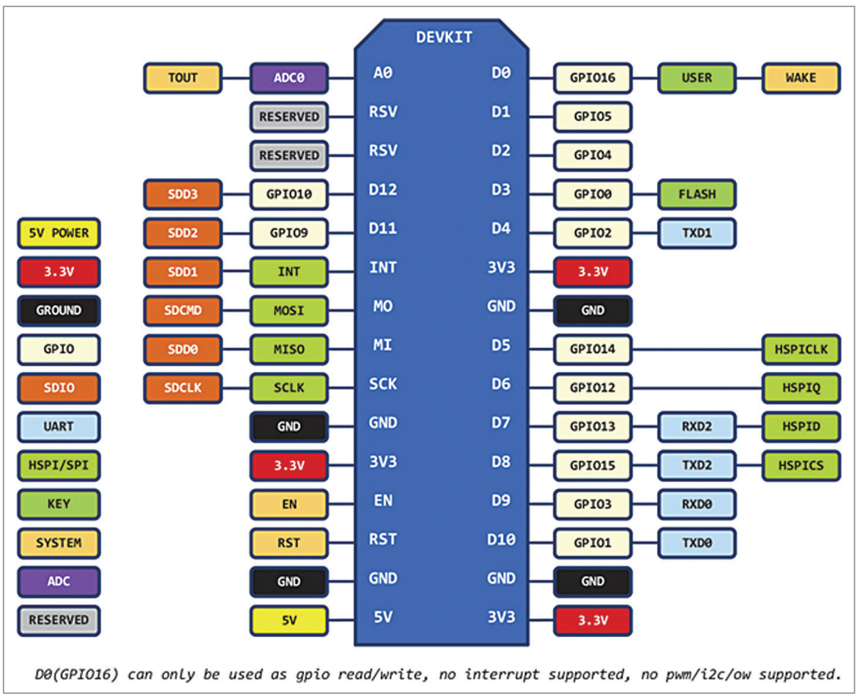


*Fig. 2: Pin diagram of NodeMCU*